Embien Technologies · Problem 4 · Transfinitte'26, NIT Tiruchirappalli

MAJoRS

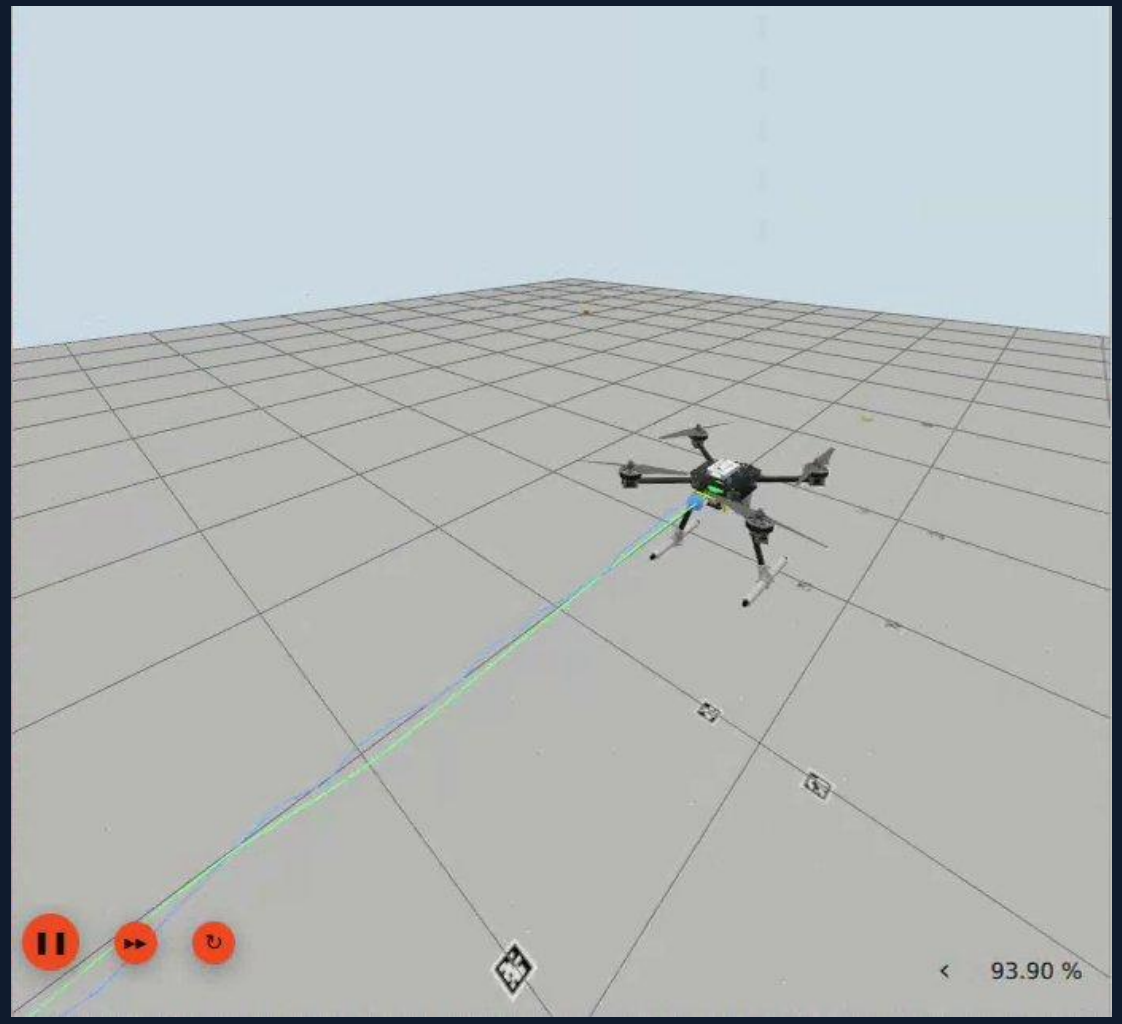
Indoor positioning with an IMU at the core

Phone IMU → measured noise model → 15-state ESKF, aided by AprilTags on real hardware and by UWB in simulation and on a public dataset

3.9 cm
RMSE · Gazebo drone + UWB

13.8 cm
RMSE · i-ASL UWB dataset

13.5 cm
end error · 33.8 m real walk



Gazebo: PX4 drone over the tag track, live ESKF path (blue) on top of ground truth (green)

One ESKF, three test beds

Overview

Phone IMU
Bosch BMI26x in a CPH2661, accel + gyro at 395 Hz

→

ADB streaming
SensorServer app: every sample, hardware timestamps

→

Calibrate + model
bias, then Allan noise N · B · K per axis

→

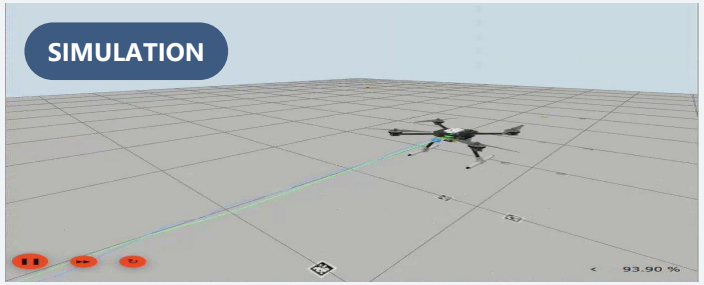
15-state ESKF
predict at IMU rate, correct with an aid

→

Position + GUI
3D trajectory, live plots, automatic reports

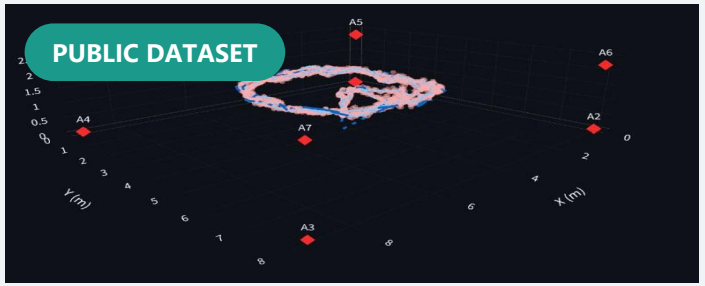
↑

Absolute aid AprilTag pose (real) · UWB ranges (sim, i-ASL) · gravity for tilt



Gazebo drone + 8 UWB anchors

3.9 cm
live RMSE vs ground truth, 1514 samples



i-ASL flight, 8 real UWB anchors

13.8 cm
3D RMSE over a 100 s flight



Handheld phone on an AprilTag track

13.5 cm
end-point error after 33.8 m

Where we stand against the brief

Problem 4 tiers

Tier	Target	What we measured	Status
1 · Core	Calibration, orientation filter, under 1 m over the first 30 s	Bias and a full Allan noise model from 1 h static. Attitude ESKF 0.4–1.3° RMSE on synthetic motion with the phone's noise. The bare IMU is 7.46 m off at 30 s, so we aid it: the aided ESKF stays under 1 m for the first 30 s of the real walk.	**Met with an aid**
2 · Main	≤ 3 % of distance on a marked 50 m path, return-to-start reported	Real walk of 33.8 m: end point 13.5 cm from the AprilTag fix (0.4 % of distance). Return-to-start: ESKF 0.95 m, camera 0.90 m (the walk ended 0.9 m from its start).	**33.8 m done · 50 m run next**
3 · Bonus	Low-cost aid keeps error under 0.3 m (or fixed-point on an MCU)	AprilTags: median 2.4 cm while tags are in view, 10.5 cm with the camera blanked half the time. UWB: 3.9 cm in Gazebo, 13.8 cm on i-ASL.	**Met with tags / UWB · no MCU port**

Aids we used, stated exactly Real walk: AprilTag fiducials + gravity. Gazebo: 8 UWB anchor ranges + gravity. i-ASL: 8 UWB anchors + magnetometer. No ZUPT, no wheel odometry, no map constraints.

How the five of us got here

Team MAJoRS

1 Whole team

Surveyed filters, from complementary and Mahony / Madgwick to EKF and ESKF, then split the work

2 Aditya Ganta

Calibration procedure for the IMU and the bias calculations

3 Joshua John

Raw phone IMU on the laptop via Developer options, ADB and the SensorServer app

4 Saiprakash

Live orientation GUI; fixed Android signal issues; static runs on all 6 channels → noise model and drift analysis

5 Rahul → Michelle

Kalman filter on synthetic Gaussian signals, then re-run with the measured Allan noise model

6 Michelle & Rahul

UWB as the correction sensor: ESKF positioning on the i-ASL dataset Michelle found

7 Joshua → Rahul

Gazebo drone carrying UWB; Rahul's live ground truth vs IMU + UWB ESKF

8 Whole team

No UWB hardware, so the same filter went onto a real phone with AprilTags as the aid

Orange: real phone and hardware · teal: filter and dataset · blue: simulation

Step 1: get every IMU sample out of the phone

Real phone data

Joshua John · Saiprakash

Polling the latest value lost about half the samples

Fix Keep every SensorEvent with its hardware timestamp

Android's calibrated streams hide the bias we want to measure

Fix Record the uncalibrated accelerometer and gyroscope

Sensors throttle in the background or with the screen off

Fix The console checks USB, Stay awake and SensorServer in front before every run

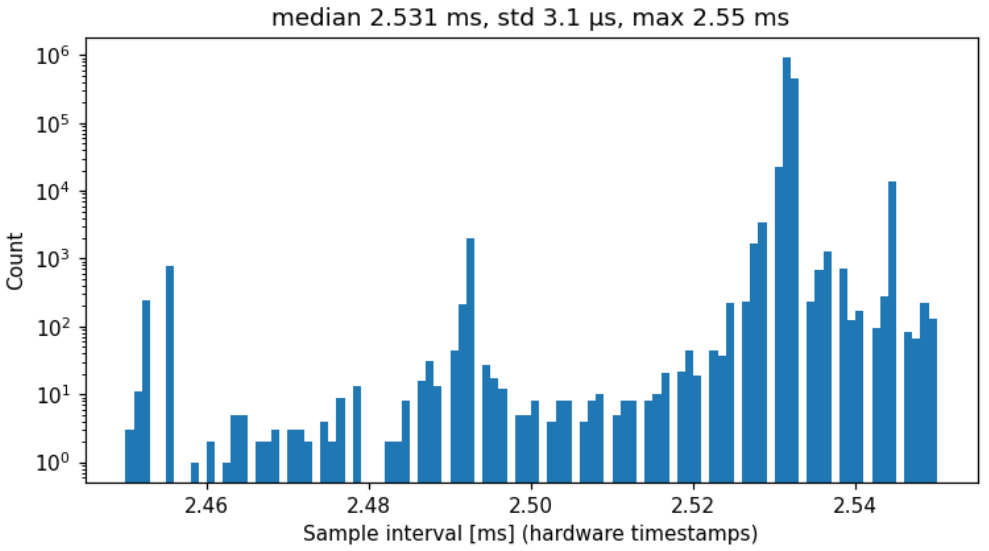
The phone does not expose the IMU temperature

Fix Log battery temperature over ADB every 10 s as a proxy

395.0 Hz
accel and gyro, same clock

3.0 µs
timing jitter (std)

0 lost
of 1,422,283 samples in 1 h



Sample intervals over the 1 h static run, from hardware timestamps

A fixed-interval assumption would add only 0.16° of heading error in that hour, so timing is not where drift comes from.

Step 2: calibrate the bias first

Real phone data Aditya Ganta

Error model (Tedaldi, Pretto, Menegatti, ICRA 2014)

$$\mathbf{a} = T_a K_a (a_S + b_a)$$
$$\boldsymbol{\omega} = T_g K_g (\omega_S + b_g)$$

T misalignment · K scale · b bias · S raw sensor reading

0.21 · −0.34 · 0.11

°/s gyro turn-on bias, x · y · z

9.839 vs 9.78 m/s²

|a| at rest vs local g: 0.06 m/s² of scale/bias

1 **Hold still, average**

Mean gyro over a still period gives the turn-on bias; the gravity direction gives initial roll and pitch.

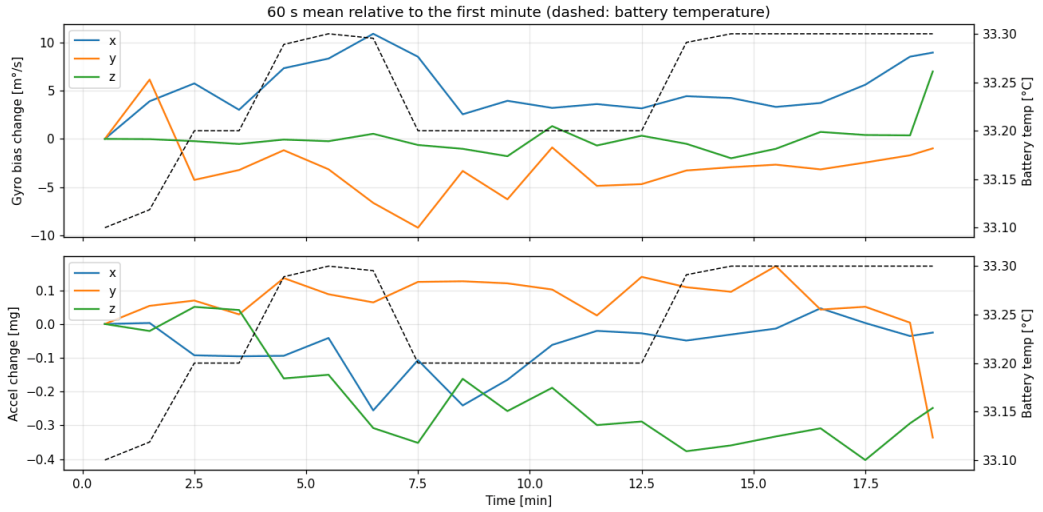
2 **Multi-position fit**

Detect static intervals, then least squares for T, K and b; the six-position method is the baseline.

3 **Align longer**

What averaging leaves behind is N / √t_align, the largest static drift term, so a longer still start cuts drift.

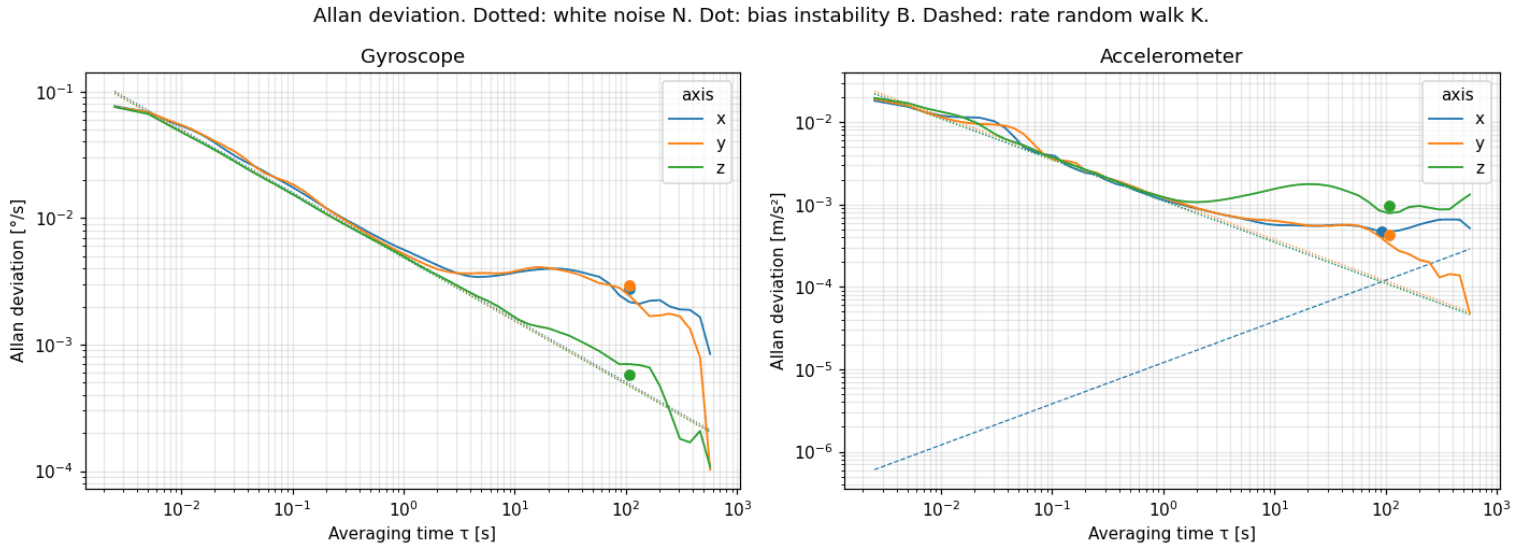
The multi-position fit is coded (nav/calib.py) and checked on synthetic data with known errors; a multi-pose run on the phone is next.



60 s mean bias over 20 min: gyro wanders ±10 m°/s and tracks the battery temperature (dashed)

One hour on a table: the phone's noise fingerprint

Real phone data · 1 h static Saiprakash



	Gyro	Accel
White noise N	4.8–5.1 m°/s/√Hz	112–121 µg/√Hz
Random walk	0.30 °/√h	0.07 m/s/√h
Bias instability B	15 °/h x, y · 3 °/h z	65–146 µg
At τ	≈ 107 s	93–107 s

−½ slope white noise: averaging helps until about 100 s

Flat bottom bias instability, the floor on how well a bias can be known

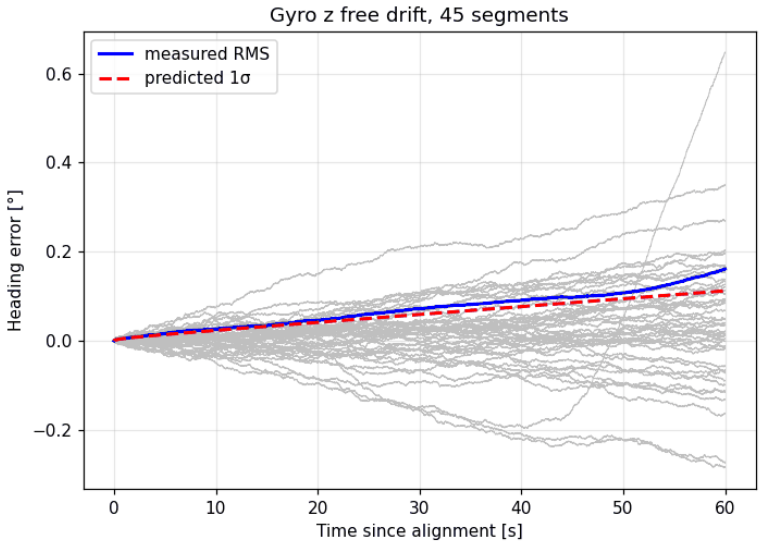
Rising tail temperature drift as the phone warmed from 33.1 to 35.6 °C

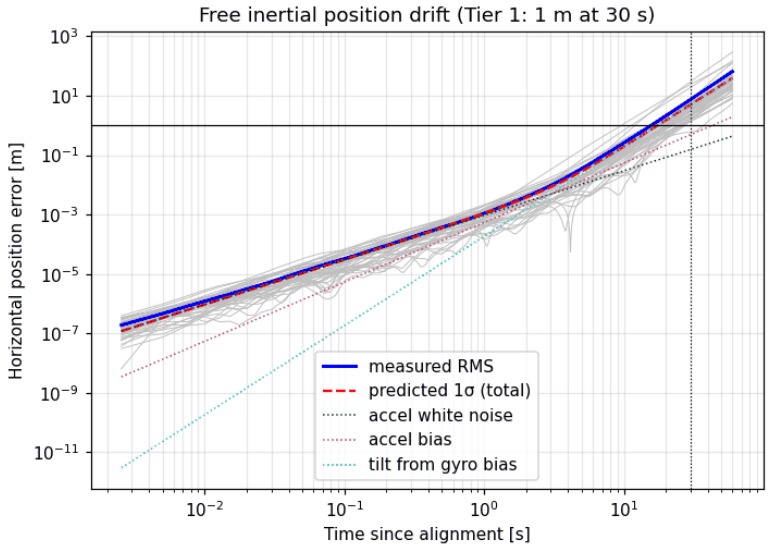
1,422,283
samples per sensor in the hour

0.69 %
of the hour disturbed (people walking past), found and cut out automatically

Unaided, the IMU is 7.5 m off after 30 s

Real phone data · 45 segments of 60 s Saiprakash





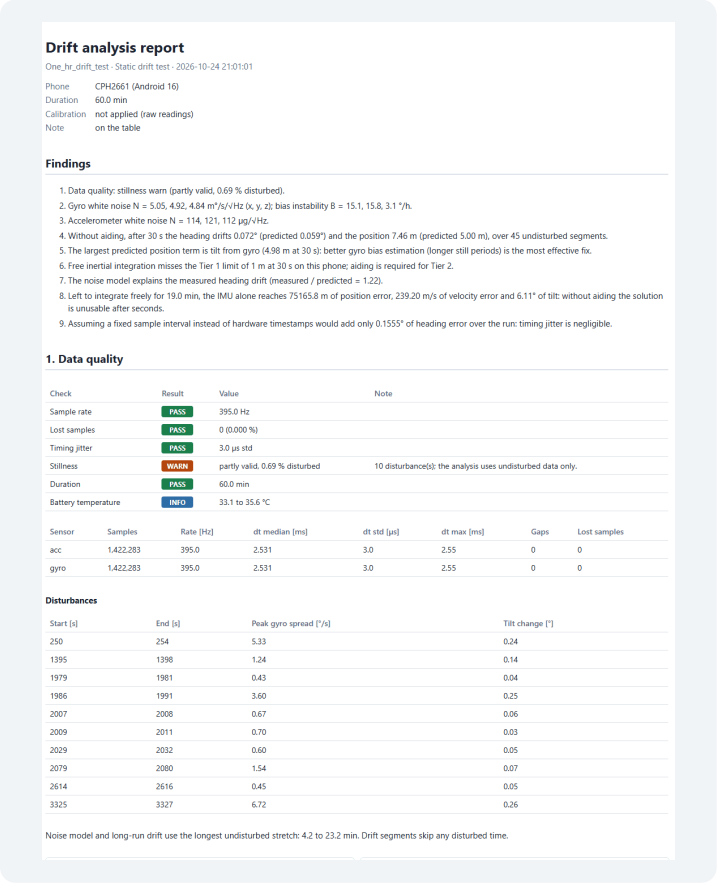
0.072°
heading at 30 s (model 0.059°)

7.46 m
position at 30 s (model 5.00 m)

4.98 m
of that from tilt by gyro bias

75 km
after 19 min free, 6.1° tilt

The model explains the drift (measured / predicted = 1.22) and names the culprit, so every test bed carries an absolute aid.



Drift analysis report

One_hr_drift_test · Static drift test · 2026-10-24 21:01:01

Phone	CPH2661 (Android 16)
Duration	60.0 min
Calibration	not applied (raw readings)
Note	on the table

Findings

1. Data quality: stillness warn (partly valid, 0.69 % disturbed).
2. Gyro white noise N = 5.05, 4.92, 4.84 m°/s/√Hz (x, y, z); bias instability B = 15.1, 15.8, 3.1 °/h.
3. Accelerometer white noise N = 114, 121, 112 µg/√Hz.
4. Without aiding, after 30 s the heading drifts 0.072° (predicted 0.059°) and the position 7.46 m (predicted 5.00 m), over 45 undisturbed segments.
5. The largest predicted position term is tilt from gyro (4.98 m at 30 s): better gyro bias estimation (longer still periods) is the most effective fix.
6. Free inertial integration misses the Tier 1 limit of 1 m at 30 s on this phone; aiding is required for Tier 2.
7. The noise model explains the measured heading drift (measured / predicted = 1.22).
8. Left to integrate freely for 19.0 min, the IMU alone reaches 75165.8 m of position error, 239.20 m/s of velocity error and 6.11° of tilt: without aiding the solution is unusable after seconds.
9. Assuming a fixed sample interval instead of hardware timestamps would add only 0.1555° of heading error over the run: timing jitter is negligible.

1. Data quality

Check	Result	Value	Note
Sample rate	PASS	395.0 Hz	
Lost samples	PASS	0 (0.000 %)	
Timing jitter	PASS	3.0 µs std	
Stillness	WARN	partly valid, 0.69 % disturbed	10 disturbance(s): the analysis uses undisturbed data only.
Duration	PASS	60.0 min	
Battery temperature	INFO	33.1 to 35.6 °C	

Sensor	Samples	Rate [Hz]	dt median [ms]	dt std [µs]	dt max [ms]	Gaps	Lost samples
acc	1,422,283	395.0	2.531	3.0	2.55	0	0
gyro	1,422,283	395.0	2.531	3.0	2.55	0	0

Disturbances

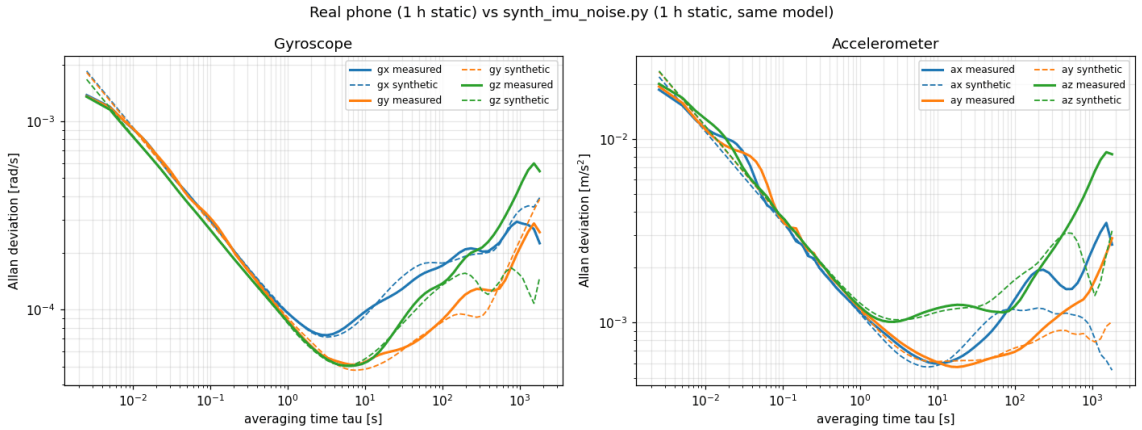
Start [s]	End [s]	Peak gyro spread [°/s]	Tilt change [°]
250	254	5.33	0.24
1395	1398	1.24	0.14
1979	1981	0.43	0.04
1986	1991	3.60	0.25
2007	2008	0.67	0.06
2009	2011	0.70	0.03
2029	2032	0.60	0.05
2079	2080	1.54	0.07
2614	2616	0.45	0.05
3325	3327	6.72	0.26

Noise model and long-run drift use the longest undisturbed stretch: 4.2 to 23.2 min. Drift segments skip any disturbed time.

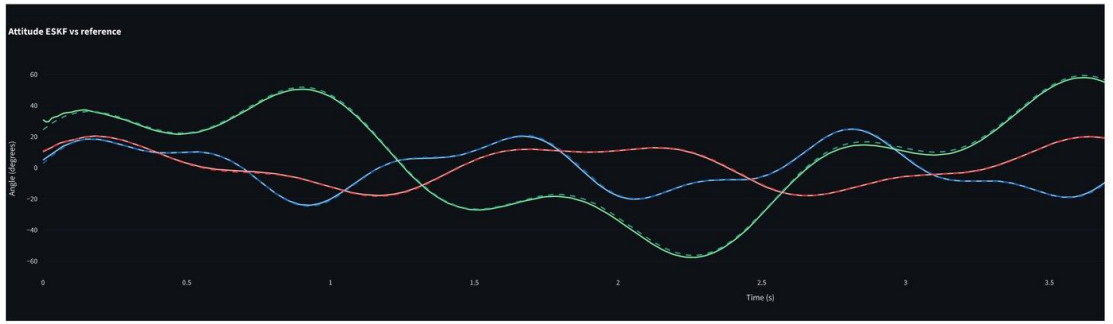
Saiprakash's console writes this report after every recording: findings, data-quality checks, noise model, drift

From Gaussian test noise to the phone's real noise

Synthetic motion + measured noise | **Rahul → Michelle**



Real phone (solid) vs our generator (dashed), 1 h each: within 2–4 % (gyro) and 5–9 % (accel) up to τ = 600 s



Attitude ESKF (solid) vs truth (dashed): roll, pitch, yaw on a synthetic 5 s manoeuvre

One noise model, three users

- Rahul's filter tests on synthetic trajectories
- Gazebo /imu/noisy at 200 Hz: gyro σ 1.25e-3 rad/s, accel σ 0.017 m/s^2
- The ESKF's Q matrix: $N^2 \cdot dt$, bias walk $\max(K^2, 2\sigma^2/\tau)$

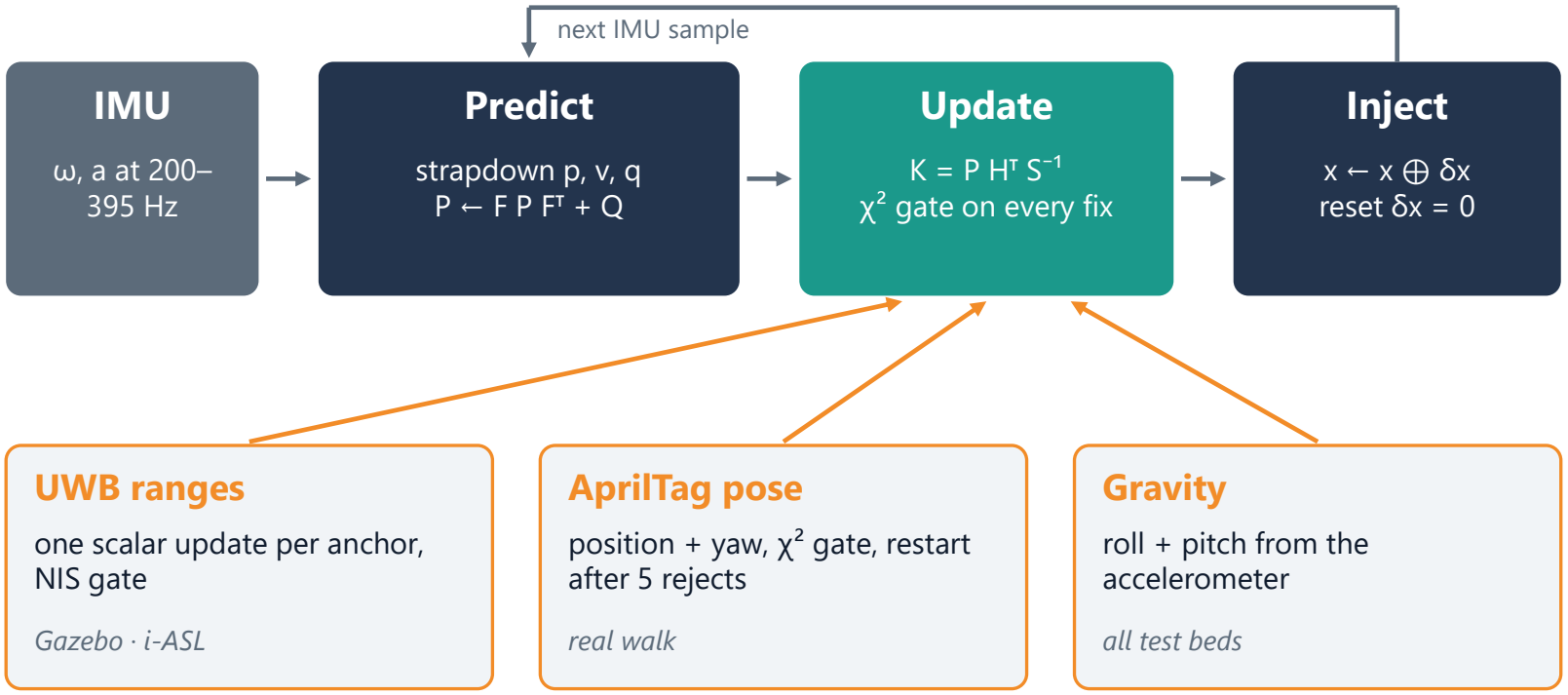
RMSE	Attitude EKF (4 states)	15-state ESKF
Roll	0.38°	0.51°
Pitch	0.35°	0.40°
Yaw	1.12°	1.34°

Auto-tune 20 Q / R pairs searched against truth; best Q = 0.001, R = 1 gives 0.70° overall. The ESKF trades a little attitude for position, velocity and bias states.

The filter: a 15-state error-state Kalman filter

Same filter in every test bed

Rahul · Michelle · Saiprakash



Error state δp, δv, δθ, δb_a, δb_g (15 states). **Q from the Allan fit** white noise $N^2 \cdot dt$; bias random walk $\max(K^2, 2\sigma^2/\tau)$, so the filter is never more sure of the bias than the sensor allows.

Why error-state

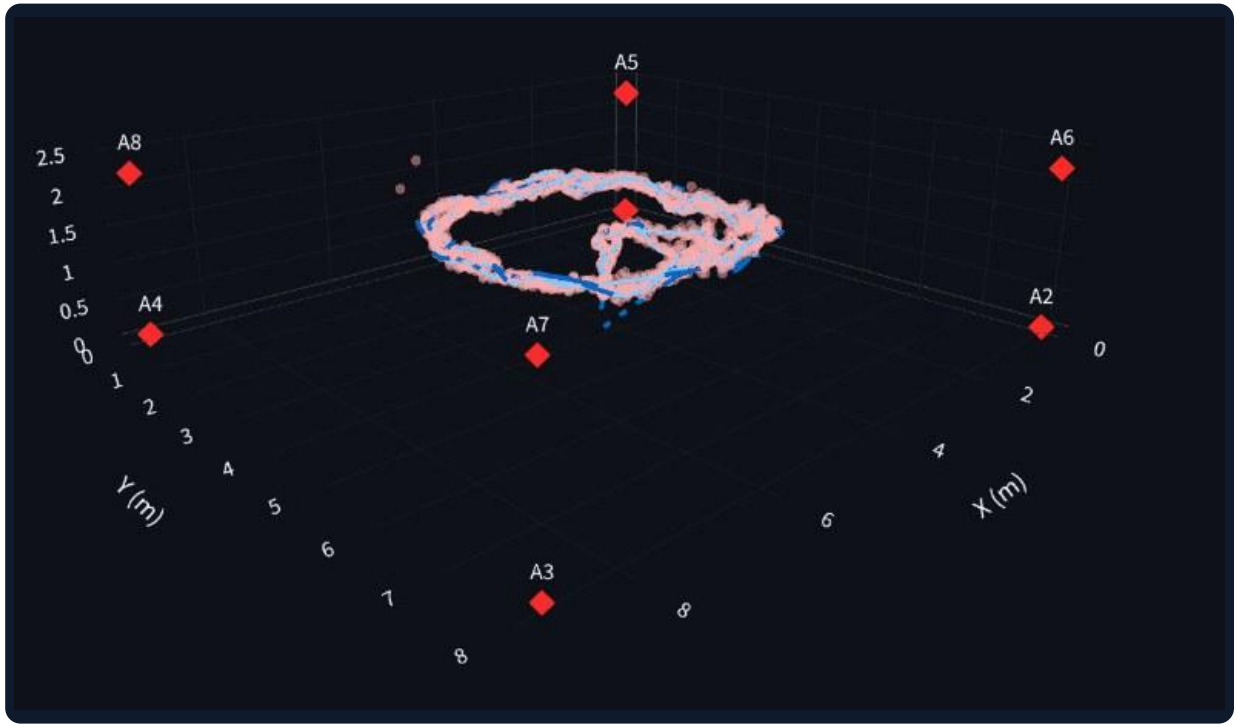
- The error stays small, so the linearisation holds through turns
- Attitude error is a 3-vector; the quaternion stays unit-norm
- Biases are states, learnt online as temperature moves them

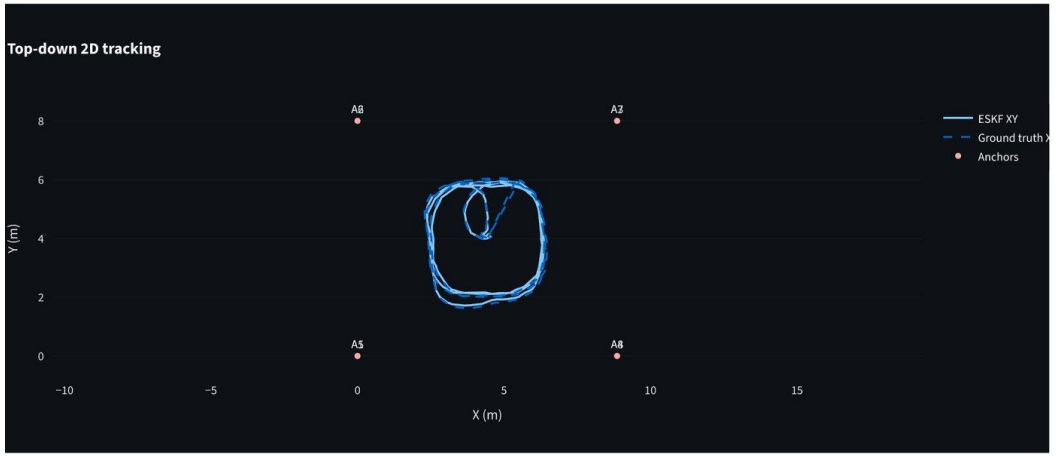
734 / 196

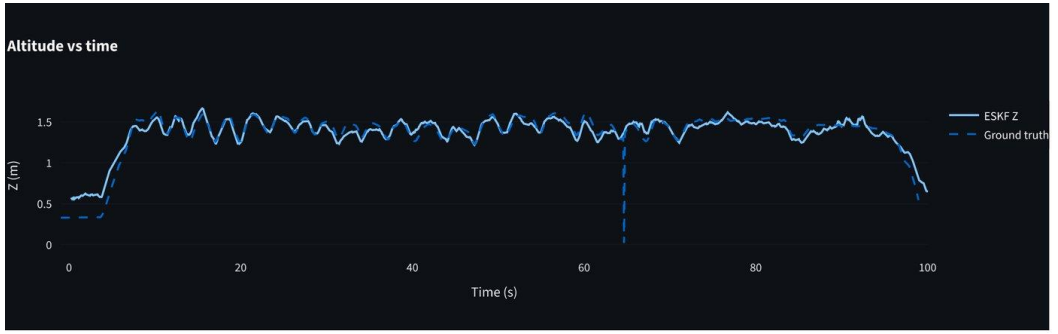
AprilTag fixes used / gated out on the real walk; 13 restarts from the camera after 5 rejected fixes in a row

Real UWB data: 13.8 cm RMSE on the i-ASL dataset

Public dataset: i-ASL (GitHub) | **Michelle · Rahul**







13.8 cm
3D position RMSE

9.5 cm
worst axis (z); x 5.6, y 8.4 cm

39,917
UWB ranges fused

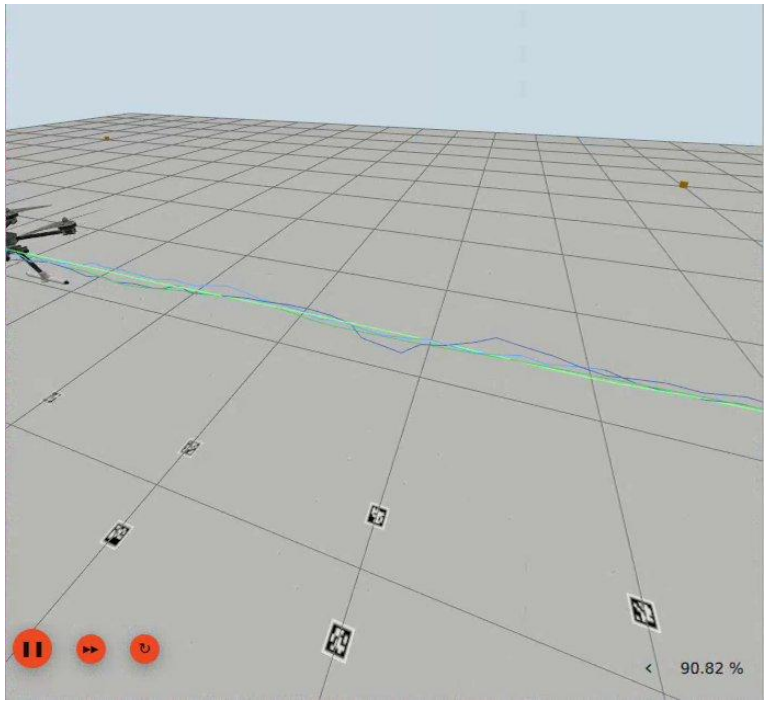
99.8 s
flight, 8 anchors

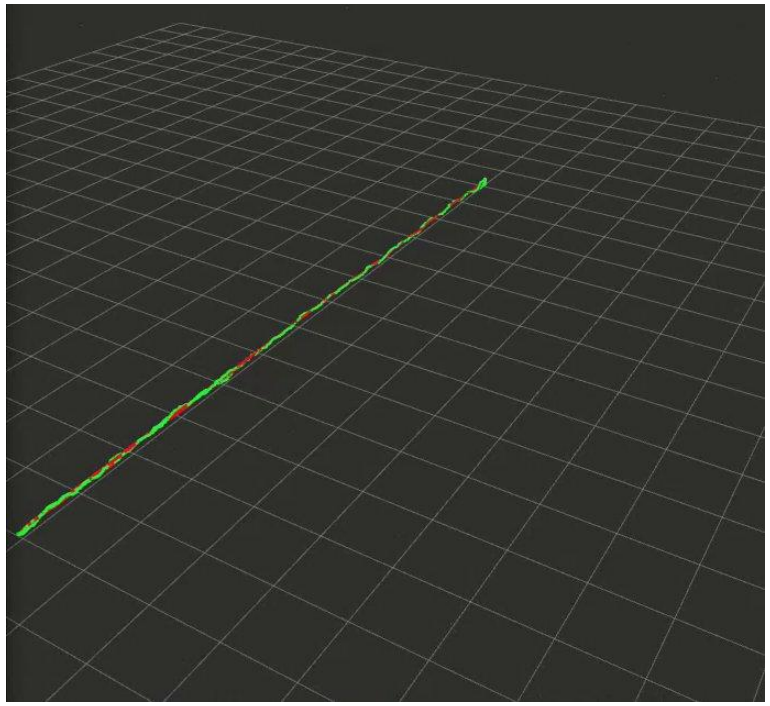
Top: ESKF (solid) on ground truth (dashed), top-down. Bottom: altitude vs time. Every anchor range is a separate scalar update; outliers are rejected by NIS, so there is no fixed 3-anchor limit.

Gazebo drone: live ESKF within 3.9 cm of truth

Simulation: PX4 SITL + Gazebo Harmonic + ROS 2 Jazzy

Joshua John → Rahul





```
[INFO] [1792902628.469902005] [eskf_live_node]: Live ESKF position RMSE vs ground truth: 0.039 m (1464 matched samples)
[INFO] [1792902630.931897042] [eskf_live_node]: Live ESKF position RMSE vs ground truth: 0.039 m (1491 matched samples)
[INFO] [1792902633.154563416] [eskf_live_node]: Live ESKF position RMSE vs ground truth: 0.039 m (1514 matched samples)
```

Left: Gazebo overlay, ESKF in blue on ground truth in green. Middle: the same run in RViz. Bottom: the node's own log; ground truth is used only for scoring.

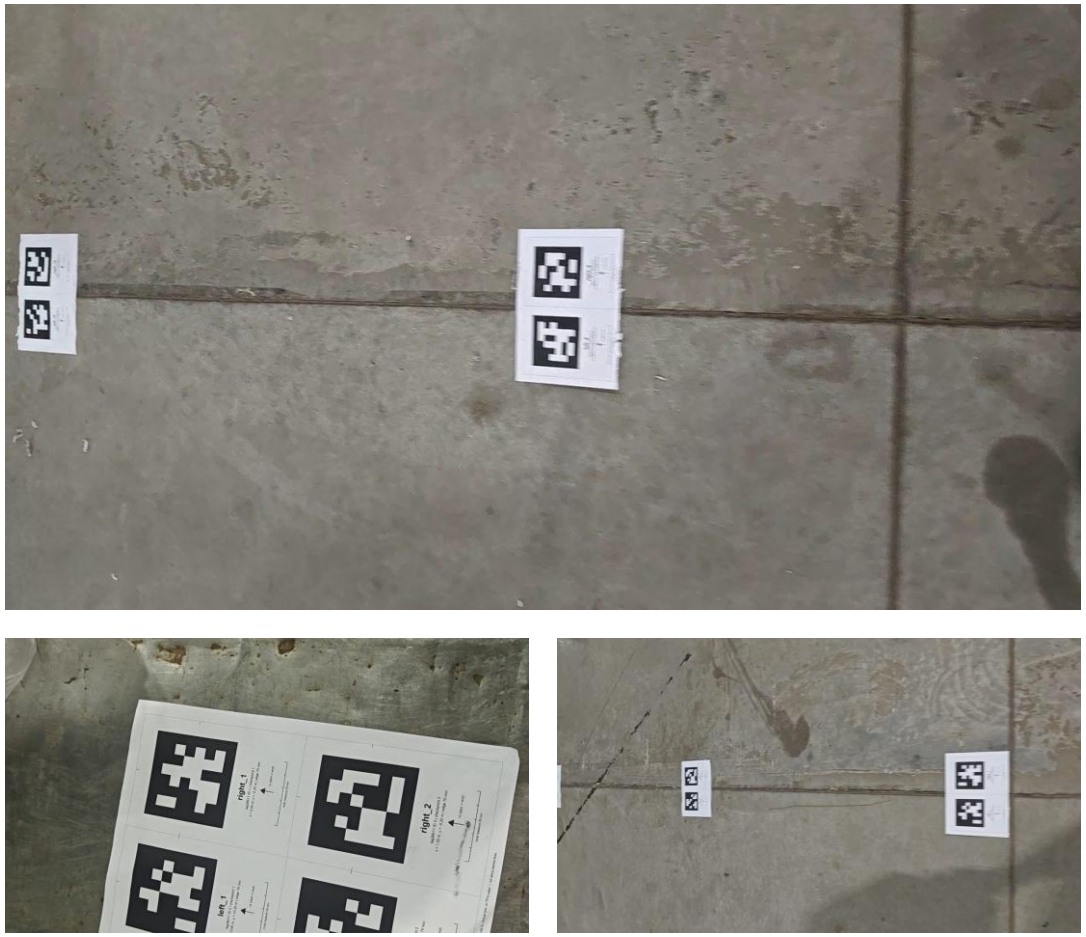
3.9 cm

live position RMSE vs ground truth, 1514 matched samples

- PX4 x500 flying the 11.5 m track at 0.3 m/s, 1.1 m up
- 8 UWB anchors on a 12 m cube, 10 Hz, σ = 5 cm
- /imu/noisy at 200 Hz with the phone's measured noise
- 22 AprilTags, same layout as our printed track
- All in Docker: Ubuntu 24.04, PX4 v1.16

No UWB hardware, so AprilTags became our anchors

Real hardware | Whole team



Frames from the phone's own camera: a checkpoint mid-walk (top), the first A4 test page and the start of the walk

1. **Track**
   11 checkpoints 1 m apart, a pair of 70 mm tag36h11 tags at each

2. **Rig**
   Phone in hand at about 1.0 m, back camera about 67° down; IMU at 395 Hz and 1080p30 video over one USB cable (record_walk.py)

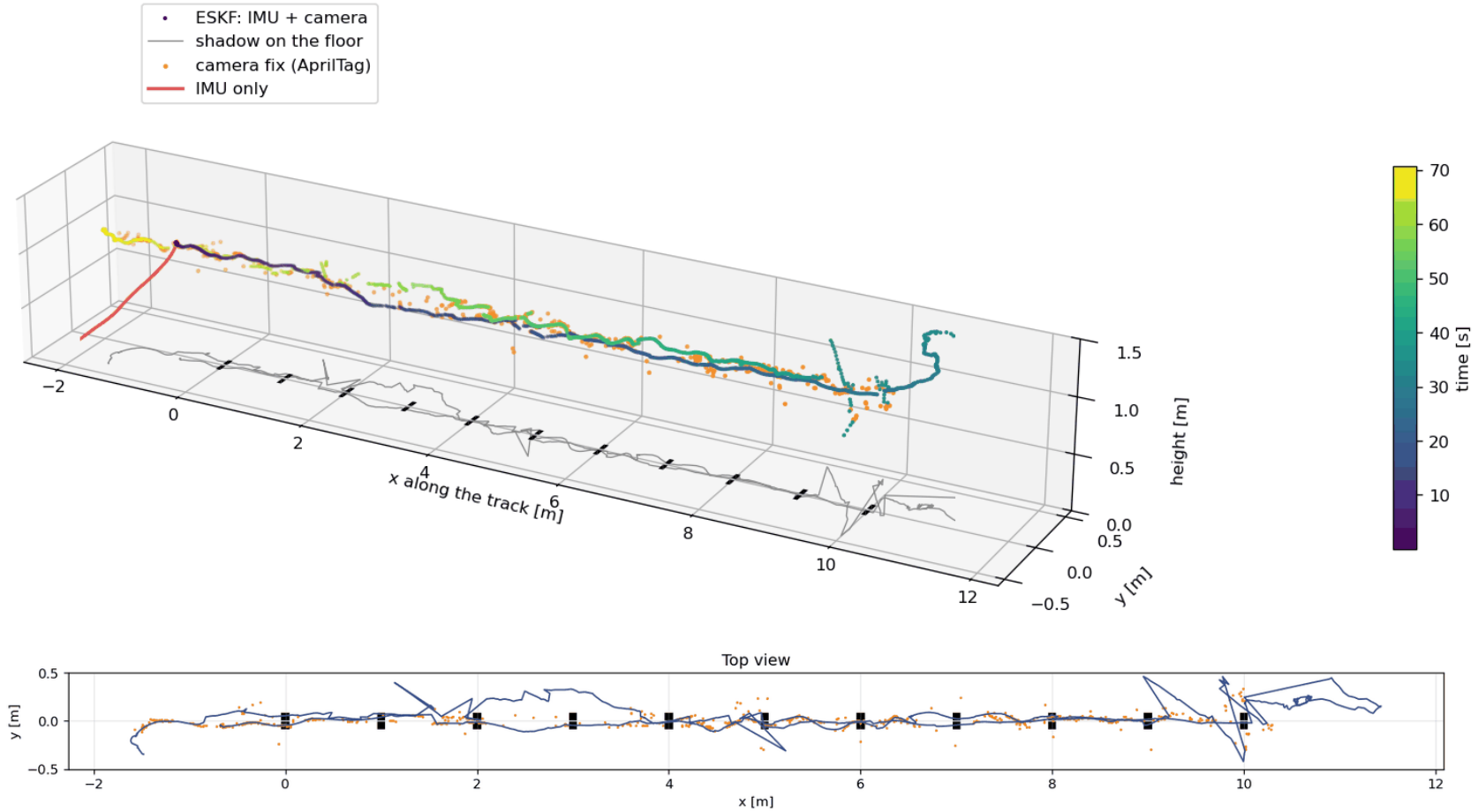
3. **Camera pose**
   OpenCV IPPE + Levenberg–Marquardt on the tag corners; fixes worse than 2 mm reprojection are dropped

4. **Sync**
   0.3 s windows of camera rotation matched to the integrated gyro: video offset −9 ms; camera-to-IMU rotation is an axis swap

5. **Why tags**
   An allowed low-cost aid that gives position and yaw, and doubles as ground truth

Real walk: 33.8 m, end point 13.5 cm off

Real hardware · walk_01 | **Saiprakash · whole team**



Top view: ESKF (blue line), AprilTag fixes (orange), checkpoints (black)

33.8 m
walked in 70.8 s

44 %
of video frames see a tag

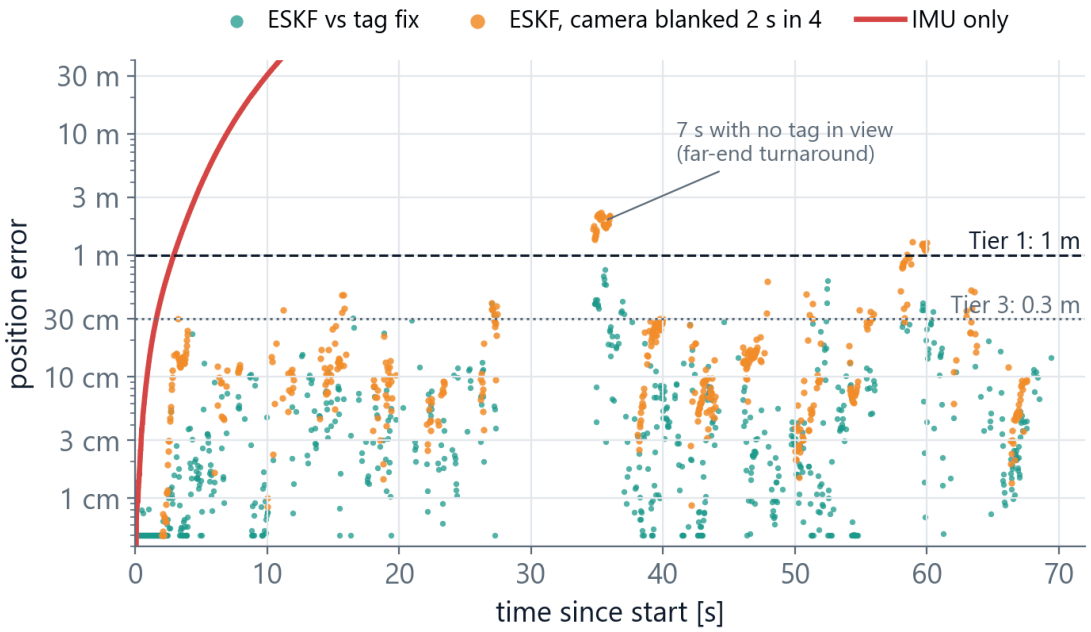
734 / 196
tag fixes used / gated out

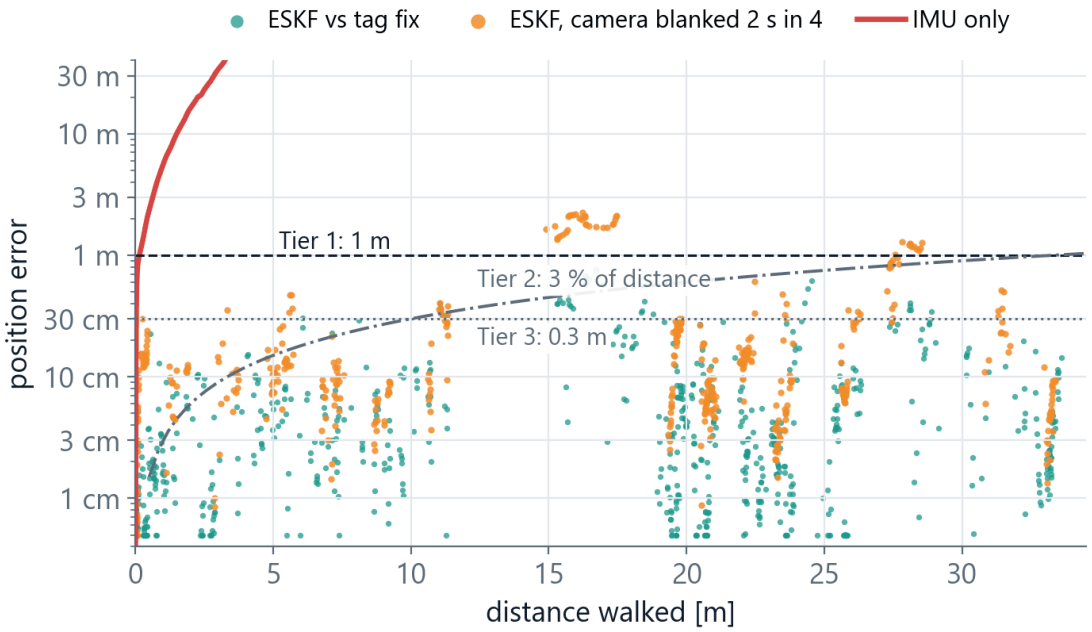
13.5 cm
end point vs the last AprilTag fix

2.9 s
until the bare IMU is 1 m off

Error vs time and distance on the real walk

Real hardware · error analysis **Computed from walk_01**





2.4 cm
median ESKF vs tag fix; 90 % under 15 cm (737 fixes)

10.5 cm
median on held-out fixes, camera blanked 2 s in every 4 s

6 → 16 cm
median bridging error, 0.5 s → 2 s after the camera drops

≈ 2 m
worst case: 7 s with no tag at the far-end turnaround

Scorecard and what comes next

35 **Accuracy**

cm-level with an aid on all three test beds: 3.9 cm sim, 13.8 cm i-ASL, 2.4 cm median on the real walk

20 **Filter and fusion**

one 15-state ESKF: per-anchor UWB, tag pose, gravity, χ^2 gating, restarts

15 **Calibration and drift**

1 h Allan noise model, drift predicted vs measured (1.22), bias procedure

15 **Real-time**

live ROS 2 ESKF on the 200 Hz sim IMU; live phone console at 395 Hz; no MCU port yet

15 **Docs and demo**

repo with README, automatic drift reports, videos, interactive 3D trajectory page

Next

- 50 m walk (5 × 10 m lengths) for Tier 2
- More tags at the turnaround to close the 7 s gap
- Checkerboard intrinsics for the phone camera
- Multi-position calibration run on the phone
- Fixed-point ESKF predict on a microcontroller

Tools Python (NumPy, SciPy, OpenCV, PyAV, Matplotlib, Plotly, Streamlit), ADB + SensorServer, scrcpy, PX4 v1.16 SITL, Gazebo Harmonic, ROS 2 Jazzy, apriltag_ros, Docker. **Data** our own phone recordings and the public i-ASL dataset.